KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering) Second Semester Examination, 2016
Second Year

ME 296 MECHANICAL ENGINEERING LABORATORY II

Answer ALL questions **TIME ALLOWED: 2 $\frac{1}{2}$Hrs**

INDEX NUMBER: **……** **DEPARTMENT: ……**

SECTION A

Instructions: Fill in the blank spaces with the appropriate word(s) for questions Q1 to Q32

Q1. The mass of a body is necessary in linear dynamics as _____ is needed in rotation dynamics.

Q2. As the acceleration, of a body increases, _____ of the same body increases.

Q3. To obtain a relationship between the _____ and temperature of saturated steam, in equilibrium with _____, at all _____ between _____ and 10 bar (150 lb/in2). • To compare between calculated values from _____ to the data obtained from _____ _____ for vfg and hfg .
.

Q4. A notch can be used to regulate the _____in open channels and rivers.

Q5. _____ manometer tubes, _____, are parts of the _____ apparatus.

Q6. _____is a device used to measure liquid _____ in a system by measuring the height to which a column of the liquid rises against gravity.

Q7. To avoid errors _____readings are taken from the _____of the meniscus.

Q8 Two parts of the Burdenberg pressure gauge tester are _____and _____

Q9. The mechanical efficiency of a machine may depends on _____advantage and _____

Q10 The process of pouring molten or liquid metal or plastic into a _____cavity for a required shape is _____.

Q11. _____involves forcing the sand into the _____ by using the rammer.

Q12. _____ and _____ are other types of casting aside sand casting.

Q13. During the process of calibration of indicator spring, the _____ is placed on the vertical _____

Q14. _____produced in industry can be used for laundry services.

Q15. _____of a machine depends on load and _____

Q16. In your experimental work you plotted the graph of temperature against absolute pressure and established the slope. What does the slope represent when analysing the data. …….…….

Q17. State the percentage of water employed when preparing sand casting mould …… ……..……..

Q18. State the percentage of clay when preparing a sand casting mould ……..

Q19. Which type of Venturi meter experimental setup was used to conduct the flow measurement experiment in the laboratory …_____

Q20. Indicate the laboratory you and your group conducted the following experiments:

i. Flow over Notches …_____..
ii. Differential pulley _____
iii. Calibration of pressure gauge and indicator spring _____ _____
iv. Flow over Wiers …_____

Q21. Indicate an application of the differential pulley experiment you conducted in the laboratory. ……

Q22. State one important activity you would need to establish in sand casting mould to have successful pouring of the molten metal …… ……

Q23. From **Q.22** indicate which other hole(s) is equally important in sand casting mould preparation. …… …… …… ……

Q24. How many ingates did your Team created for your sand casting mould?
……

Q25. In Marcet boiler experiment, the relationship between pressure and temperature is found to be directly proportional. True or False? …….

Q26 The reading of increase and decrease in temperature during the experiment was to calculate to …….

Q27. How can the industry apply corrections to the gauge readings during calibration of pressure gauge experiments? ……..…….
……..

Q28. From the experiment in **Q27** more oil was used in the experiment to ……...

Q29 From the experiment in **Q27** the ……. of the plunger was stop the moment the piston ……..

Q30 The meter used as flow sensing element in the industries and has no projections into the fluid, no sharp corners, and no sudden changes in contour measurement is known as ……

Q31 In your flow measurement experiment you quoted the equation below in your theory

$$Q = C_d \cdot A_2 \sqrt{\frac{1}{1-\beta^4}} \sqrt{\frac{2(P_1 - P_2)}{\rho}}$$

Explain the ***beta factor (β) in the equation*** **……**

……..……

Q32a Which flow measurement meter used in your experiment at the fluid mechanics laboratory would you employ when dealing with range of flow rates ……

b) Give reasons of your choice in **Q32a**
……
……
……

<u>Underline</u> the appropriate responses to the following questions Q33 – Q43 on the question paper.

Q33 What are the dimensions of orifice coefficient?

a) kg.m/s
b) m/s
c) dimensionless
d) m^2/s

Q34 If the flow rate increases then the orifice coefficient

a) Decreases
b) Increases
c) Remains constant
d) None of the above

Q35 The typical value of orifice coefficient is

a) 0.975
b) 0.5
c) 0.7
d) 0.6

Q36 The typical value of venturi coefficient is

a) 0.975
b) 0.7
c) 0.5
d) 0.6

Q37 Orifice meter can be used to measure flow rates of

a) Gases only
b) Liquids and gases
c) Liquids only
d) Solids, liquids and gases

Q38 The Vena contracta is the

a) point of maximum convergence
b) point of maximum flow
c) point of maximum expansion
d) none of the above

Q39 Select the most appropriate condition of pressure and velocity of a fluid flowing across an orifice plate:

I. The pressure in the upstream is less and the velocity is more at downstream
II. The pressure in the upstream is equal to the pressure in the downstream
III. At upstream, both the velocity of the fluid and the pressure will be more to at the downstream
IV. after the orifice plate the fluid particles will try to regain their energy back
V. after the orifice plate the velocity will be less and the pressure will be more

Q40 Select the appropriate device you would use to find discharge of rain water

a) Rotameter
b) Venturimeter
c) Orifice meter
d) barometer

Q41. Underline the TRUE statement below. The time you allowed for the air to escape from the boiler during the experiment on the relationship between the temperature and pressure of a saturated steam was to:

a. Allow for saturation of the steam
b. Deal with the inaccuracy in the experiment
c. Follow manufacture's specification
d. Minimise the difference between the graphs of pressure against dT/dP
e. Minimise the difference between the graphs of absolute pressure and Tv_g/h_{fg}

Q42. Underline the correct statement. Marcet boiler can be applied to:

i. Power plant
ii. Cooking utilities
iii. Water boiler
iv. Gasifier
v. i, ii, iv & iii
vi. i, iii & iv only

Q43. Which of the following activities did NOT contribute to the small deviation between experimental slope and theoretical slope?

a. Steam was allowed to come out from the vent valve for about 30seconds before the valve was closed.
b. Air may be not have completely flowed out from the valve so the reading was not being constantly increased.
c. Besides the (b) above, may be temperature was taken while the pressure was still not stable.
d. Typical human error in reading the values due to parallax

Section B

Answer ALL questions in this section. Provide the appropriate answer to the questions in the answer booklet and where applicable on a graph sheet.

Q1. From the calibration of pressure gauge and indicator spring experiment conducted Table 1 was recorded.

i. plot graph of appropriate pressure against the indicator distance

ii. From the graph of **Q1(i)** establish the slope

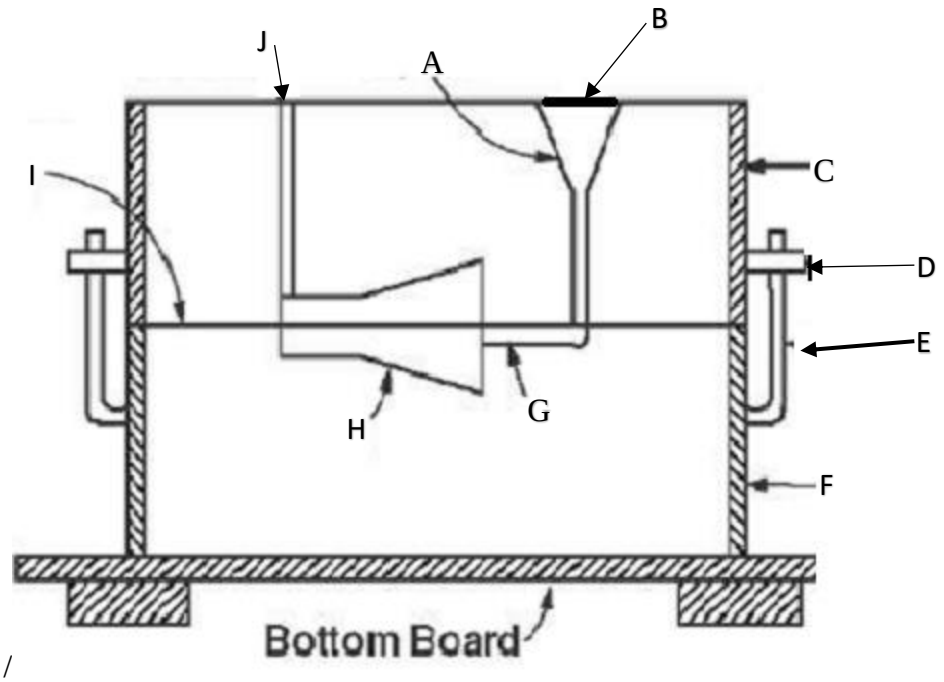
iii. From **Q1(ii)** state the experimental value for the spring constant.

iv. From **Q1 (iii)** indicate the deviation in kN /m^2/m and in percentage (%) if the printed value on the spring is **2.72 ×10kN/m^2/m**

Table 1: Experimental data of calibration of pressure gauge and indicator spring

TRUE PRESSURE (KN/m^2)	GAUGE READING (KN/m^2)	INDICATOR DISTANCE (m)
69	96.6	0.005
138	158.7	0.007
207	234.6	0.01
276	303.6	0.012
345	365.7	0.015
414	434.7	0.018
483	510.6	0.02
552	572.7	0.022
621	648.6	0.025
690	710.7	0.027
759	786.6	0.03
828	855.6	0.032
897	924.6	0.035
966	993.6	0.037
1035	1062.6	0.039
1104	1138.5	0.043
1173	1207.5	0.045
1242	1276.5	0.047

Q2a Label Figure Q2a below



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Figure Q2a

Q2b State the three essential parts of sand casting mould:

i. _____
ii. _____
iii. _____

Q3a. State the name and label the experimental setup shown in Figure Q3a

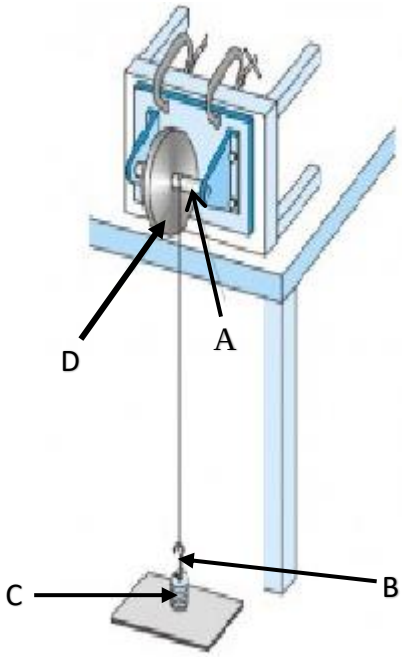


Figure Q3a: _____

Q3b What do you understand by Moment of Inertia (MOI) ?

Q3c: Give the expression for the MOI of Flywheel.

Q3d. List the apparatus required in the experiment of Q3a

I.A. Edwin